WASTE AND HAZARDOUS WASTE MANAGEMENT

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MINI PROJECT REPORT

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1. Introduction

Plastic bottle waste, primarily made of Polyethylene Terephthalate (PET) and High-Density Polyethylene (HDPE), continues to accumulate in large volumes due to widespread single-use consumption and limited recycling infrastructure. This project proposes a practical method of converting such waste into a useful construction material: an outdoor brick or paver formed by crushing plastic bottles into small pellets and binding them together with a resin. The resulting composite brick offers a low-cost, durable, and environmentally beneficial alternative to conventional clay or concrete bricks for outdoor use such as pathways, garden borders, and paved surfaces.

2. Problem Statement

Conventional plastic waste disposal methods such as landfilling and incineration contribute to soil degradation, water contamination, and air pollution. At the same time, the construction industry continues to rely heavily on resource-intensive materials such as fired clay bricks and cement concrete, both of which have significant carbon footprints. There is a clear need for a process that simultaneously reduces plastic waste volume and produces a functional, reusable building material.

3. Objectives

- To develop a simple, replicable process for converting waste plastic bottles into construction-grade pellets.
- To bind the crushed plastic pellets using a resin matrix to form a solid, moulded brick.
- To evaluate the brick's suitability for outdoor applications based on strength, water resistance, and durability.
- To demonstrate a low-cost, small-scale recycling model that can be replicated at community or institutional level.

4. Materials Required

Material / Equipment	Purpose
Waste PET/HDPE bottles	Raw material for plastic pellets
Plastic shredder / crusher	Reduces bottles into uniform-sized pellets/flakes
Sieve set	Grades pellets by size for uniform packing
Epoxy or polyester resin	Binder that holds pellets together
Hardener / catalyst	Initiates curing of the resin
Mould (brick-shaped)	Gives shape to the final brick
Release agent	Prevents brick from sticking to mould
Vibration table (optional)	Removes trapped air, improves compaction

5. Process Methodology

The fabrication process consists of twelve sequential stages, from raw bottle collection to final quality testing, as illustrated in the flowchart below.

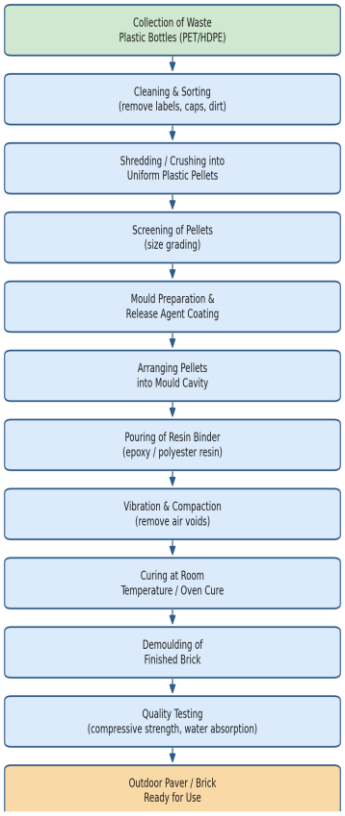


Fig. 1 — Process flow for plastic pellet–resin brick fabrication

5.1 Stage Description

- Collection & Cleaning: Bottles are collected, washed, and stripped of labels/caps to remove contaminants.
- Shredding: Bottles are fed into a crusher/shredder to produce small plastic pellets or flakes (roughly 3–8 mm).
- Screening: Pellets are passed through sieves to separate oversized or undersized fragments.
- Mould Preparation: The brick mould is cleaned and coated with a release agent for easy demoulding.
- Pellet Arrangement & Resin Pouring: Graded pellets are packed into the mould, then resin mixed with hardener is poured over them to fill the voids.
- Compaction & Curing: The mould is vibrated to remove trapped air, then left to cure at room temperature or in an oven until fully hardened.
- Demoulding & Testing: The cured brick is removed and tested for strength and water absorption before use.

6. Brick Composition Diagram

The finished brick is a composite material — a solid resin matrix surrounding compacted plastic pellets. The diagram below shows the internal composition of the cured brick, including the pellet distribution and the layered resin structure.

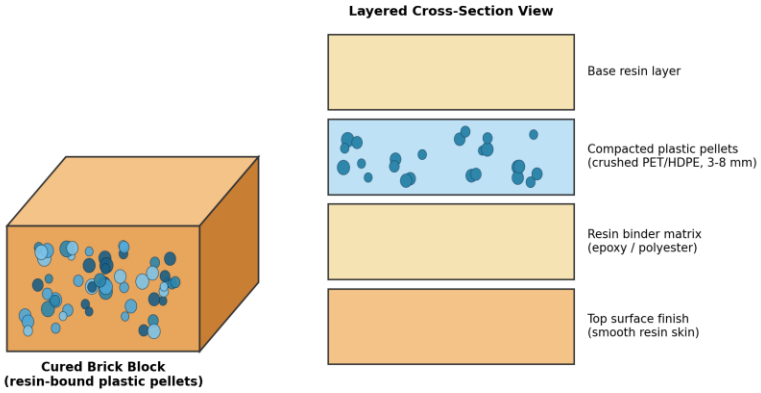


Fig. 2 — Internal composition of the plastic pellet–resin brick

7. Advantages & Limitations

7.1 Advantages

- Diverts single-use plastic waste from landfills and open dumping.
- Lightweight compared to conventional concrete or clay bricks, with resin coating giving good water resistance for outdoor exposure.
- Simple, low-cost process that can be set up at small or community scale, reducing demand for clay/cement.

7.2 Limitations & Considerations

- Resin cost and availability can affect overall production cost.
- Long-term UV exposure may affect surface appearance of the resin coating.
- Load-bearing use requires proper compressive strength testing before scaling up.
- Fumes from resin curing require adequate ventilation and safety precautions.

8. Applications

- Garden and landscape pathways
- Outdoor paving and courtyard flooring
- Boundary/edging bricks for lawns and flower beds
- Low-load decorative outdoor structures

9. Conclusion

Converting crushed plastic bottle pellets into resin-bound outdoor bricks presents a practical and scalable method to address plastic waste while producing a genuinely useful construction material. The process is straightforward, requires modest equipment, and results in a lightweight, water-resistant brick suitable for outdoor, non-structural applications. With further strength testing and process refinement, this method has strong potential as a small-scale sustainable manufacturing model.

10. Final Product

